

Polarization Accurate Simulations of Complex Multilayer Optical Films for Automotive Display Systems

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Abstract

Accurate simulation of polarization is essential for many display applications. Birefringent and coherent effects make simulating polarization in ray tracers difficult. To address this, we have developed a method that uses Mueller matrix data integrated with commercial ray tracing software. Validation was performed using a simple cross-polarizer model, followed two automotive case studies that utilize reflective polarizers.

Author Keywords

HUD; Heads-up display; Polarization; Cold Mirror; Speos; LightTools; Simulation; Automotive display; Reflective Polarizer; Thermal Management.

1. Introduction

For the past few decades, traditional non-sequential ray tracing has been used to simulate the performance of optical systems. This includes things like LCD backlights, projectors, illuminations systems, automotive lighting and heads-up displays. While still a powerful tool, traditional ray tracing treats light as discrete rays, and does not account for the wave nature of light. This limitation has prevented accurate simulations of multilayer optical films where thin film interference is the basis for their optical properties. This limitation is especially relevant for reflective polarizer films which use thin film interference with many layers of birefringent materials to achieve their unique optical properties [1].

Commercial ray tracing software packages have attempted to solve this limitation by implementing several methods of approximating film behavior. The most common method is to use tabulated data that contains the transmission and reflection coefficients for the s, and p polarizations states at various angles of incidence (polar angle) and wavelength [2, 3]. This method can be effective for simple thin films such as anti-reflective coatings but fails to accurately model complex multilayer films such as reflective polarizers or circular polarizers.

To overcome these difficulties there are two options that immediately present themselves. First, one could use a secondary physics engine in real-time as rays interact with the surface to calculate the full wave physics of the film and pass the information back to the ray tracer. In this work, we have implemented a second method where the polarization characteristics of the film are tabulated in a more thorough manner using a Mueller matrix formalism. Assuming sufficient sampling within the tabulated data including angle of incidence (polar angle), plane of incidence (azimuthal angle), wavelength, front/back orientation the ray tracer can accurately simulate the behavior of complex multilayer films such as reflective polarizers.

2. Background Information

Mueller Matrix Generation: Mueller matrices are generated using a custom proprietary thin film calculation software

developed at 3M. Though there are now commercial options available that perform similar calculations [4, 5]. The software uses the transfer matrix method to calculate the full Mueller matrix for a given multilayer film stack at specified angles of incidence (polar and azimuthal), wavelength, and incident side.

LightTools: A user defined optical property (UDOP) has been created that allows LightTools to read in the Mueller Matrix data files. It also performs the necessary calculations to take in the incident ray information and calculate the transmitted and reflection rays. The UDOP also allows the user to specify the orientation of the film. It also allows both an option for an in-plane rotation, or bias, and an option for flipping the construction to swap the front and back sides of the film. This is important for films that are not symmetric in construction and have different optical properties when illuminated from the front or back sides and when there are different media on either side of the film.

Speos: For Speos, a plugin surface optical property (SOP) has been developed that allows Speos to read in the Mueller Matrix data files. Similar to the LightTools UDOP, the SOP performs the necessary calculations to take in the incident ray information and calculate the transmitted and reflection rays.

3. Comparison vs built-in methods

A simple setup has been designed to test the Mueller matrix method in each software. See this example in Figure 1, where planar surfaces were created and one was rotated 90 degrees. Linear polarization models were added to each surface to create a cross-polarizer model. A conoscopic intensity sensor was added to the model to measure the transmitted light.

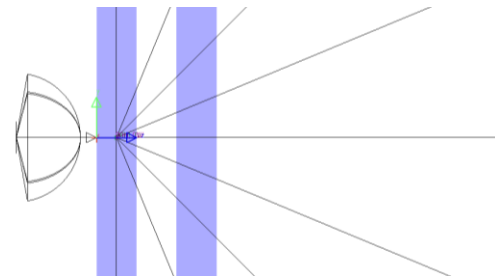


Figure 1. Schematic of cross-polarizer system in LightTools.

In this system the cross-polarizers were simulated in two ways. First, a coating optical property was used that implemented the coating as a tabulation of t_s , t_p vs wavelength and polar angle. Second, the UDOP that used tabulated Mueller matrix was used. Results can be viewed in figure 2. In this figure it can be observed that the coating method does not recreate the expected Maltese cross pattern [6]. While there are other built-in methods within the software that are all useful in their own right, they all have limitations that prevent them from matching the UDOP. The other methods that are available are not under examination in this paper.

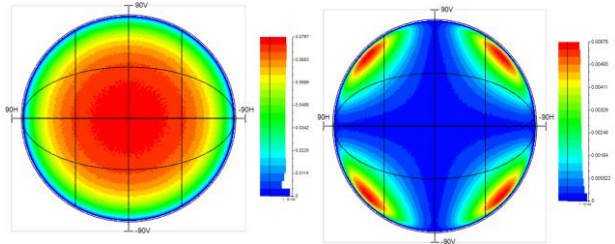


Figure 2. Output of cross-polarizer model. a) shows the output using an optical coating and b) shows the output using the UDOP.

4. Case Study: Head-up Display (HUD) in Speos

A Heads-up Display (HUD) system is shown in Figure 3, where s-polarized light from the picture generation unit (PGU) is projected to the windshield surface. To eliminate ghosting due to reflections off the inner and outer windshield surface, the windshield is approximated to a single interface. To reduce solar heat radiation reaching the PGU, a 3M™ Cold Mirror Film (CMF) is used as the fold mirror in the HUD system. The CMF is a multilayer reflective polarizer which reflects over 95% of the polarized visible light from the PGU and transmits 50% of the visible solar radiation and over 90% of the near-IR solar radiation.

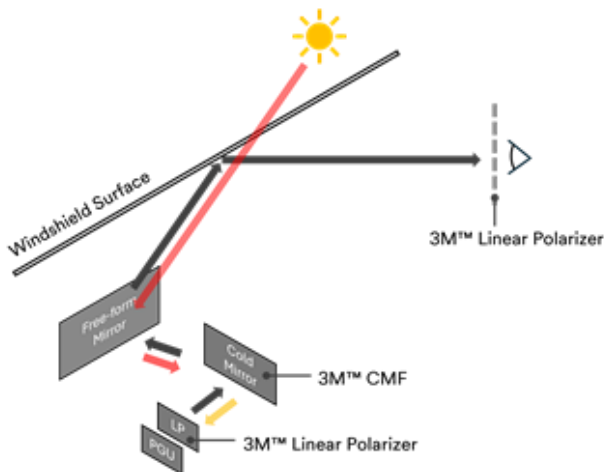
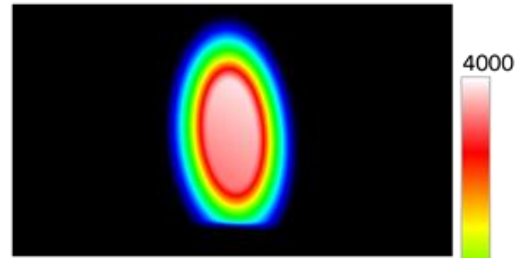


Figure 3. Schematic of the HUD system.

To estimate solar energy reaching PGU and demonstrate 3M CMF as a potential fold-mirror replacement, we performed optical simulation of the HUD system with sun as the illumination source. Solar irradiance maps on the PGU for a 100% reflective cold mirror and 3M CMF are shown in Figure 4. Incoming solar power was adjusted to obtain 50,000 Lux on the outer surface of the windshield. For the 100% reflective cold mirror, the entire spectral band of the solar radiation is directed towards the PGU surface, measuring a flux of 1.2W. By replacing the cold mirror with 3M™ CMF, the flux on PGU lowers to 0.14W, which is about 88% reduction.

(a) Irradiance on PGU (with Cold Mirror)



(b) Irradiance on PGU (with 3M™ CMF)

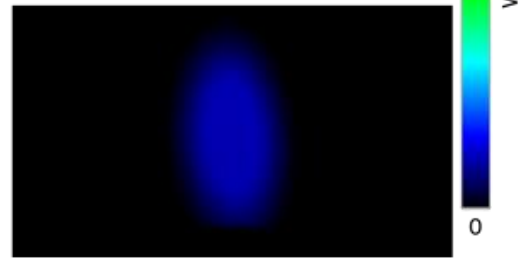
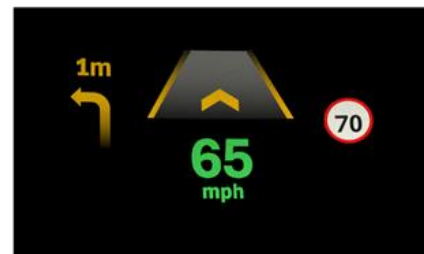


Figure 4. Solar irradiance on PGU (Speos Simulation).

A typical luminance map of HUD image is shown in Figure 5. The PGU was set to emit 0.7 Lumen prior to the linear polarizer and generating about 3000 Nits maximum luminance in the HUD image zone, which is within the acceptable range of HUD luminance under daylight ambient lighting (Maximum luminance of 10,000 Nits) [7]. With the addition of a polarized sunglass, the HUD luminance drops by 84% as is evident in Figure 4(b).

(a) Luminance (S-pol HUD Image)



(b) Luminance (S-pol HUD Image with Polarized Sunglass)

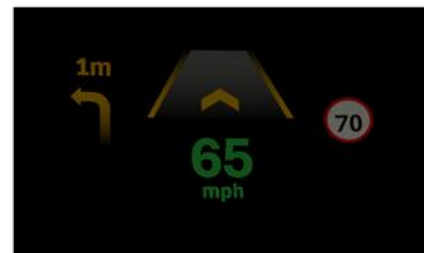


Figure 5. Image of the projected HUD image without and with polarized sunglasses.

5. Case Study: Black Print Head-up Display Backlight in LightTools

A windshield black print head up display system including the backlight is shown in figure 6, where there is a back reflector, a light guide, a 3M™ Brightness Enhancement Film (BEF), a 3M™ Image Redirecting Film – Reflective Polarizer (IDF-RP), and an LCD panel. The IDF-RP has a prism layer and a reflective polarizer layer. After leaving the LCD panel the light will reflect off the windshield or other element of the vehicle into the eyebox of the viewer. The IDF-RP film serves a dual purpose; it has a reflective polarizer to help recycle light in the backlight and a prism layer to turn the output light distribution to a desirable angle for reflection into the eyebox of the driver.

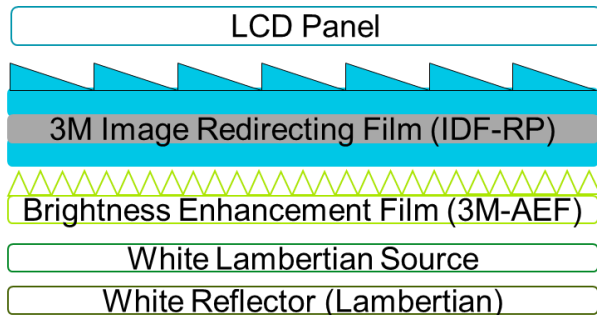


Figure 6. Schematic of IDF-RP backlight system

A model of the backlight system was built in LightTools. In Figure 7 the angular output of the backlight with and without the IDF-RP. The effect of the prismatic layer on the light output is clear, turning the peak from on-axis to and an angle of 5.5 degrees which is better suited for reflection to the observer.

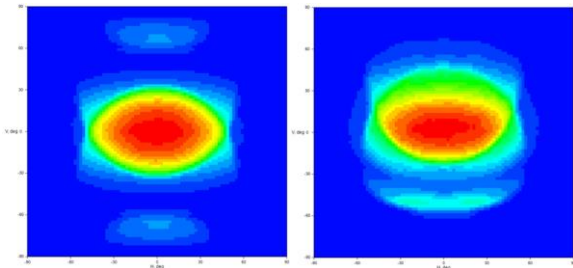


Figure 7. Conoscopic output of the black print head up display with and without Image Redirecting Film – Reflective Polarizer (IDF-RP). Left shows the angular output without IDF-RP and right shows the tilted output with it.

Examining the data further the peak brightness of the output has been calculated. In the case where the IDF-RP is present the peak value is 40% higher than the case without the IDF-RP.

6. Conclusion

Through both case studies we have shown that it is now possible to accurately simulate complex polarizers including reflective polarizers in commercial ray trace software. This was

accomplished using new custom written plugins for LightTools and Speos. Using the new capability we have shown that you can simulate two different types of automotive heads-up displays and capture important phenomena such as polarized sun glass readability, backlight recycling, and solar flux on interior pieces.

There are other applications where this capability could be useful. Anywhere polarizers are used could be imagined. Extended reality devices often use reflected polarizer in their components, like pancake lenses, and polarizing beam splitters. In the consumer electronics spaces the new capabilities could be used to optimize the circular polarizer in an OLED display for example.

7. Impact of Research

The UDOP plugin is available now on the 3M™ Digital Materials Hub (<https://digitalmaterialshub.3m.com/>). The installer for the plugin can be found on the learning center page for optical models and then the individual optical model files can be downloaded from the product catalog.

8. Acknowledgements

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9. References

1. Weber MF, Stover CA, Gilbert LR, Nevitt TJ, Ouderkirk AJ. Giant birefringent optics in multilayer polymer mirrors. *Science*. 2000;287(5462):2451-2456. doi:10.1126/science.287.5462.2451
2. LightTools Illumination Design Software | Keysight. Accessed November 25, 2025. <https://www.keysight.com/us/en/products/software/optical-solutions-software/optical-design-solutions/lighttools.html>.
3. Speos coating surface format – Ansys Optics. Accessed November 25, 2025. <https://optics.ansys.com/hc/en-us/articles/30422704374931-Speos-Coating-Surface-Format>.
4. STACK Optical Solver Overview. Ansys Optics. Published 2022. Accessed November 25, 2025. <https://optics.ansys.com/hc/en-us/articles/360034914653-STACK-Optical-Solver-Overview>
5. Insideoptics.com. Published 2025. Accessed November 25, 2025. <http://www.insideoptics.com/wp/>
6. Hecht, Eugene. Optics, 2nd ed., Addison Wesley (1990) ISBN 0-201-11609-X. Chapter 8.
7. SAE International Technical Standard, Standard - Optical System HUD for Automotive, SAE Standard J1757-2_201811, Issued November 2018, https://doi.org/10.4271/J1757-2_201811.